



Write a MATLAB function that clamps elements of an input vector to a specified range

Clamping refers to restricting a value to a specified range: values outside the range are truncated to the nearest bound, while values within the range remain unchanged. Write a MATLAB function that accepts 3 inputs:

- 1. a vector x of K elements,
- 2. a lower bound, ld (a scalar),
- 3. and an upper bound, lu (a scalar).

The function should clamp all elements of x to the range of $[ld, lu]$. Assume $ld < lu$. Specifically:

$$y(x)_i = \begin{cases} ld, & \text{if } x_i < ld \\ lu, & \text{if } x_i > lu \\ x_i, & \text{otherwise} \end{cases} \quad \text{for } i = 1, \dots, K \text{ and } x = (x_1, x_2, \dots, x_K) \in \mathbb{R}^K.$$

Function ?

Save

Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

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```
1 function y = clamp(x, ld, lu)
2 % The function definition is provided for you.
3
4 %% your code goes here
5 y = zeros(1, length(x));
6 for i = 1:length(x)
7     y(i) = min(max(x(i), ld), lu);
8 end
9
10 %%
11
12 end
13
```

Code to call your function ?

Reset

```
1 % define lower and upper limit
2 ld = -6; % example lower limit
3 lu = 4; % example upper limit
4
5 % example x vector
6 x = -12:2:12;
7
8 % calling your function
9 clamp(x, ld, lu)
```

Run Function



Tests: All Tests Passed

Submit



Are the values correct